

Def.

A deformation retraction of a space  $X$  onto a subspace  $A$  is a family

$\{f_t: X \rightarrow X \mid t \in I\}$  satisfying  $f_0 = \text{id}$ ,  $f_1[X] = A$ ,  $f_t|_A = \text{id}_A$  for all  $t$ ,  
and  $X \times I \rightarrow X: (x, t) \mapsto f_t(x)$  is continuous.